Forward semantics

$$s, \texttt{skip} \Downarrow^+ s, \epsilon, \epsilon \qquad s, \langle c \mid \gamma \rangle \Downarrow^+ cs, [s], [\gamma] \qquad \frac{s, C_1 \Downarrow^+ s', \sigma, \Gamma \quad s', C_2 \Downarrow^+ s'', \sigma', \Gamma'}{s, C_1 \,;\, C_2 \Downarrow^+ s'', \sigma' :: \sigma, \Gamma' :: \Gamma}$$

$$\frac{\phi(s) \quad s, C_1 \Downarrow^+ s', \sigma, \Gamma}{s, \texttt{if}\ \phi\ \texttt{then}\ C_1\ \texttt{else}\ C_2 \Downarrow^+ s', \sigma, \Gamma} \qquad \frac{\neg\phi(s) \quad s, C_2 \Downarrow^+ s', \sigma, \Gamma}{s, \texttt{if}\ \phi\ \texttt{then}\ C_1\ \texttt{else}\ C_2 \Downarrow^+ s', \sigma, \Gamma}$$

$$\frac{\phi(s) \quad s, C \Downarrow^+ s', \sigma, \Gamma \quad s' \prec s \quad s', \texttt{while}_\prec\ \phi\ \texttt{do}\ C \Downarrow^+ s'', \sigma', \Gamma'}{s, \texttt{while}_\prec\ \phi\ \texttt{do}\ C \Downarrow^+ s'', \sigma' :: \sigma, \Gamma' :: \Gamma} \qquad \frac{\neg\phi(s)}{s, \texttt{while}_\prec\ \phi\ \texttt{do}\ C \Downarrow^+ s, \epsilon, \epsilon}$$

Backward semantics

$$\sigma, \Gamma, t \Downarrow^- \sigma, \Gamma, t \qquad s :: \sigma, \gamma :: \Gamma, t \Downarrow^- \sigma, \Gamma, \gamma s t \qquad \frac{\sigma, \Gamma, t \Downarrow^- \sigma', \Gamma', t' \quad \sigma', \Gamma', t' \Downarrow^- \sigma'', \Gamma'', t''}{\sigma, \Gamma, t \Downarrow^- \sigma'', \Gamma'', t''}$$